Wildlife Conservation Donation Analysis: Evaluating Donor Behavior, Campaign Effectiveness, and Revenue Stability in Nonprofit Fundraising

1. Executive Summary

Wildlife conservation charities depend heavily on donations to sustain their programs, yet fundraising performance is often influenced by complex interactions between donor demographics, engagement strategies, and campaign effectiveness. Without structured analysis, it becomes difficult to identify which factors drive revenue and where growth opportunities exist.

This project analyzes 10,000 donation records from a simulated wildlife conservation dataset to uncover key drivers of fundraising performance. The analysis integrates donor segmentation, campaign evaluation, channel effectiveness, and time-based trends to provide a comprehensive view of donation behavior.

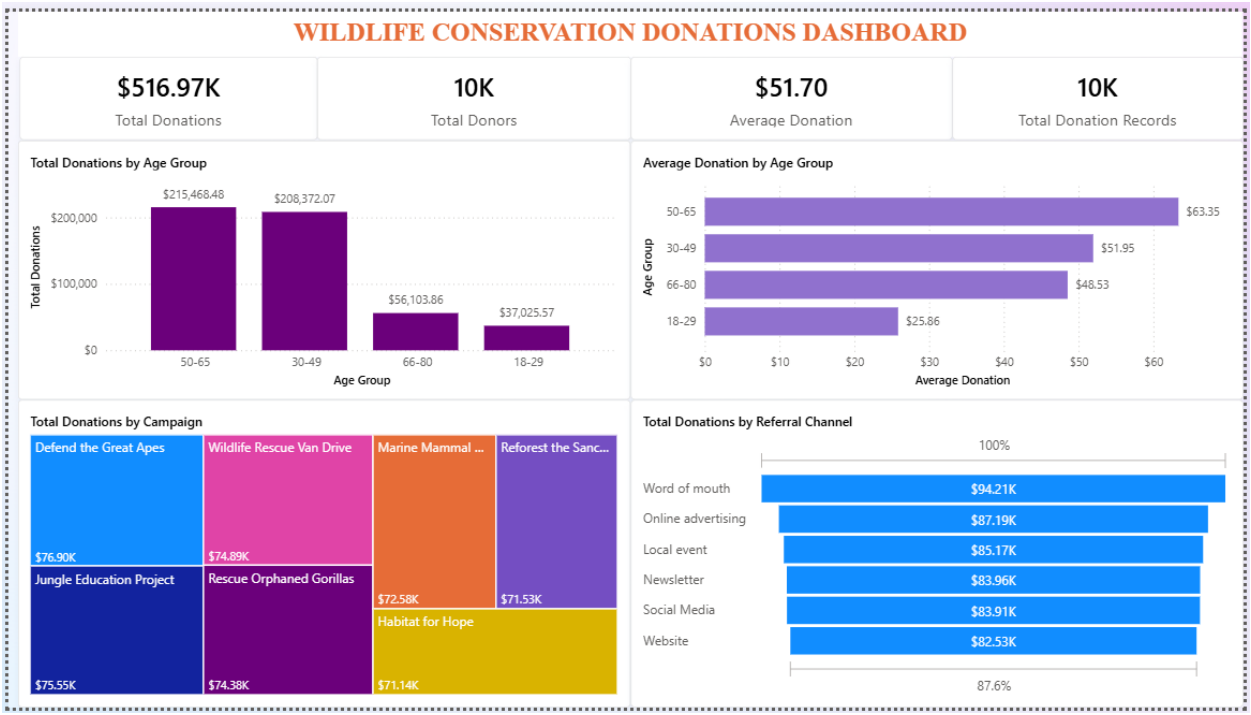
The findings reveal a clear pattern: donation revenue is largely driven by older donor segments and high-performing acquisition channels, while engagement mechanisms such as newsletters show limited impact on conversion. Additionally, the charity relies heavily on one-time donations, creating revenue volatility.

Forecasting results indicate stable but flat donation trends, suggesting that while donor retention is strong, growth remains limited without strategic intervention.

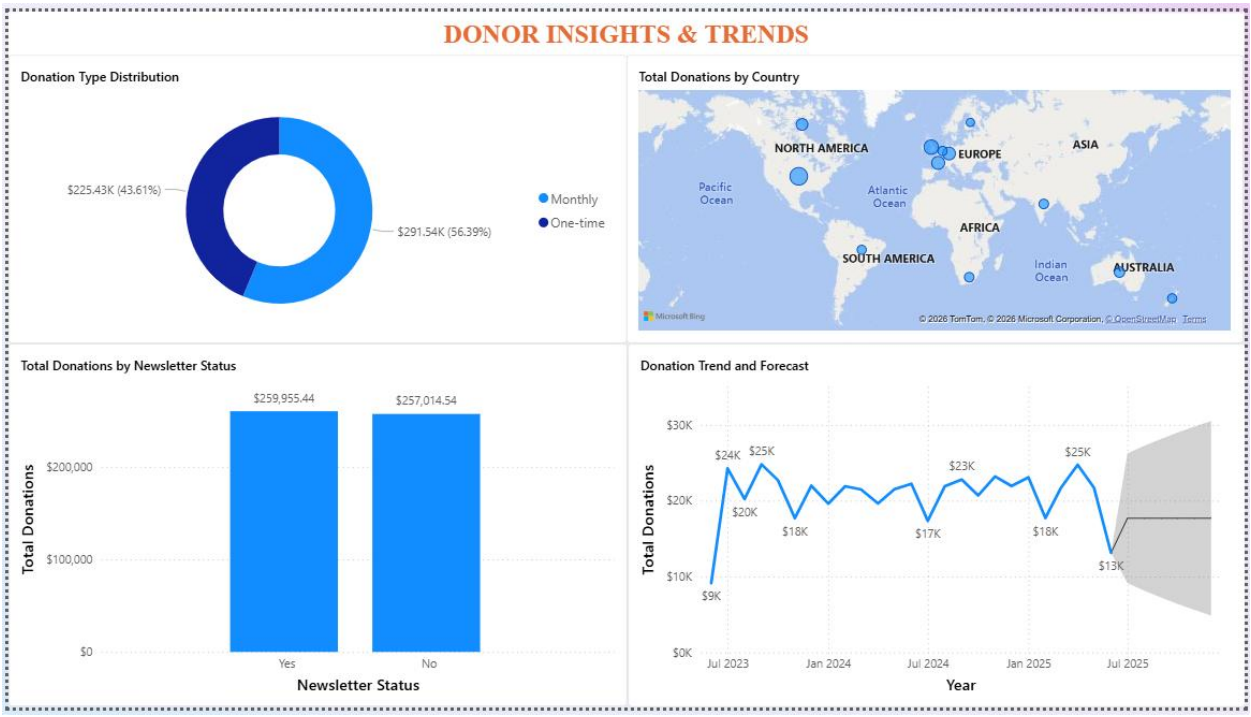
Next Steps:

- Transition toward recurring donation models to improve revenue stability
- Optimize high-performing channels such as word-of-mouth and digital acquisition
- Redesign engagement strategies to improve conversion
- Introduce growth-focused fundraising initiatives

2. Dashboard Preview



Provides a high-level summary of total donations, donor segments, campaign performance, and referral channel effectiveness.



Explores donation type distribution, geographic contribution patterns, engagement impact, and time-based trends with forecasting.

3. Business Problem

Wildlife conservation charities collect donations across multiple campaigns, channels, and donor segments, but often lack visibility into the underlying drivers of fundraising performance.

Key challenges include:

- Identifying which donor segments contribute the most value
- Understanding which campaigns and channels drive revenue
- Evaluating whether engagement strategies influence donation behavior
- Monitoring how donation trends evolve over time
- Anticipating future performance for strategic planning

Without clear insights into these relationships, fundraising decisions remain reactive, limiting the organization's ability to optimize revenue, allocate resources effectively, and scale its impact.

This project aims to address these challenges by analyzing donation data to uncover actionable insights that support data-driven fundraising strategies.

4. Methodology

This analysis combines multiple analytical approaches to examine donation behavior across donor segments, channels, and time.

- Data Cleaning & Transformation
  Used Power Query to standardize data types, validate consistency, and prepare the dataset for analysis
- Feature Engineering
  Created time-based variables (Year, Month), donation segmentation categories, and engagement labels to enable deeper analysis
- Exploratory Data Analysis (EDA)
  Analyzed distributions across donor demographics, donation types, campaigns, and channels
- Segmentation Analysis
  Identified high-value donor groups based on age and contribution patterns
- Channel & Funnel Analysis
  Evaluated referral channels to determine acquisition effectiveness
- Time Series & Forecasting
  Assessed monthly donation trends and applied forecasting to project future performance

5. Skills

Power BI

- Data modeling and relationships

- DAX (measures, aggregations, calculated columns)
- Power Query (data cleaning and transformation)
- Dashboard design and storytelling
- Forecasting (Analytics pane)

Data Analysis

- Exploratory Data Analysis (EDA)
- Segmentation analysis
- Time series analysis
- Performance benchmarking

Analytical Thinking

- Translating data into business insights
- Identifying behavioral patterns
- Structuring end-to-end analytical narratives

6. Key Results and Business Recommendations

6.1. Key Insights

- Donor Segmentation
  Older donors (ages 50–65) contribute the highest total and average donation values, indicating a strong concentration of financial capacity within this group.
- Donation Behavior
  One-time donations account for the majority of contributions, highlighting reliance on irregular revenue streams and limited adoption of recurring giving.
- Channel Effectiveness
  Word-of-mouth and online advertising generate the highest donation values, suggesting these channels are the most effective in acquiring high-quality donors.
- Campaign Performance
  Campaign contributions are relatively balanced, indicating a diversified fundraising strategy but also a lack of clear standout campaigns.
- Engagement Analysis
  Newsletter subscribers and non-subscribers contribute similar donation levels, suggesting that current engagement strategies are not effectively driving conversion.
- Trend & Forecast
  Donation trends remain stable over time with moderate fluctuations. Forecasting indicates limited growth, suggesting strong retention but weak expansion.

6.2. Business Recommendations

- Improve Revenue Stability
  Introduce recurring donation programs to reduce reliance on one-time contributions and create predictable income streams.

- Target High-Value Donors
  Develop tailored campaigns for the 50–65 age segment, focusing on high-impact messaging and long-term engagement strategies.
- Optimize Acquisition Channels
  Reallocate resources toward high-performing channels such as word-of-mouth and digital advertising to improve return on investment.
- Enhance Engagement Strategy
  Redesign newsletter content to focus on conversion through segmentation, personalization, and stronger calls-to-action.
- Scale High-Performing Campaigns
  Identify key drivers of successful campaigns and replicate these strategies to improve overall fundraising effectiveness.
- Drive Growth Initiatives
  Introduce new acquisition strategies, partnerships, and geographic expansion to move beyond stable performance and achieve growth.

7. Next Steps

- Incorporate donor-level behavioral data to enable lifetime value (LTV) analysis
- Conduct A/B testing on campaigns and engagement strategies
- Integrate external factors (seasonality, economic trends) into forecasting
- Develop predictive models for donor conversion and retention
- Expand geographic analysis for more targeted fundraising strategies